Modelling

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```
if (!require("pacman")) install.packages("pacman")
```

```
Loading required package: pacman
```

```
pacman::p_load(stats, dynlm, lmtest, sandwich, urca, quantmod, exuber, readxl, exuberdata)

load("allData.RData")

y <- log(price) - log(dividend)
#this step is necessary or radf() will return error
y_vec <- ts(as.numeric(y), start = c(1926, 3), frequency = 12)

est_y <- radf(y_vec, lag = 1)
summary(est_y)
```

```
Using `exuberdata::radf_crit2` for `cv`.
```

```
-- Summary (minw = 74, lag = 1) ------------------- Monte Carlo (nrep = 2000) --

series1 :
# A tibble: 3 x 5
  stat   tstat   `90`   `95`  `99`
  <fct>  <dbl>  <dbl>  <dbl> <dbl>
1 adf   -1.85  -0.451 -0.139 0.541
2 sadf   0.984  1.33   1.59  2.10
3 gsadf  2.45   2.15   2.34  2.78
```

```r
#selecting lags
adf_model_y <- ur.df(y_vec, type = "drift", selectlags = "AIC")
summary(adf_model_y)
```

```
###############################################
# Augmented Dickey-Fuller Test Unit Root Test #
###############################################

Test regression drift


Call:
lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)

Residuals:
     Min       1Q   Median       3Q      Max
-0.28318 -0.01937  0.00372  0.02231  0.41647

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.017109   0.008859   1.931   0.0537 .
z.lag.1     -0.004722   0.002555  -1.848   0.0648 .
z.diff.lag   0.293812   0.027678  10.615   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.04341 on 1196 degrees of freedom
Multiple R-squared:  0.08711,   Adjusted R-squared:  0.08558
F-statistic: 57.06 on 2 and 1196 DF,  p-value: < 2.2e-16


Value of test-statistic is: -1.8483 1.9659

Critical values for test statistics:
      1pct  5pct 10pct
tau2 -3.43 -2.86 -2.57
phi1  6.43  4.59  3.78
```

```r
# suggests one lag
```

```
# Reject null at 5%, evidence of explosiveness
diagnostics(est_y)
```

```
Using `exuberdata::radf_crit2` for `cv`.
```

```
-- Diagnostics (option = gsadf) --------------------------------- Monte Carlo --

series1:  Rejects H0 at the 5% significance level
```

```
#Implementing rule-of-thumb proposed by Phillips et. al (2015) to compute minimum duration of
rot = psy_ds(length(y), rule = 1, delta = 1)

#trying to date periods of exuberance
dstamp_y <- datestamp(est_y, min_duration = rot)
```

```
Using `exuberdata::radf_crit2` for `cv`.
```

```
Warning: Argument 'min_duration' excludes all explosive periods
```

```
dstamp_y
# no period from the 2020s makes this criteria

y2 <- ts(as.numeric(log_price), start = c(1926,3), frequency = 12)

adf_model_y2 <- ur.df(y2, type = "drift", selectlags = "AIC")
summary(adf_model_y2)
```

```
###############################################
# Augmented Dickey-Fuller Test Unit Root Test #
###############################################

Test regression drift


Call:
lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)
```

```
Residuals:
     Min       1Q   Median       3Q      Max
-0.28074 -0.01802  0.00382  0.02280  0.39134

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.0023875  0.0033958   0.703    0.482
z.lag.1     0.0002972  0.0006343   0.468    0.640
z.diff.lag  0.2699129  0.0278759   9.683   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.04254 on 1196 degrees of freedom
Multiple R-squared:  0.07338,   Adjusted R-squared:  0.07183
F-statistic: 47.35 on 2 and 1196 DF,  p-value: < 2.2e-16


Value of test-statistic is: 0.4685 4.9982

Critical values for test statistics:
      1pct  5pct 10pct
tau2 -3.43 -2.86 -2.57
phi1  6.43  4.59  3.78
```

```
est_y2 <- radf(y2, lag = 1)
summary(est_y2)
```

```
Using `exuberdata::radf_crit2` for `cv`.


-- Summary (minw = 74, lag = 1) ------------------- Monte Carlo (nrep = 2000) --

series1 :
# A tibble: 3 x 5
  stat  tstat   `90`   `95`  `99`
  <fct> <dbl>  <dbl>  <dbl> <dbl>
1 adf   0.468 -0.451 -0.139 0.541
2 sadf  0.973  1.33   1.59  2.10
3 gsadf 2.79   2.15   2.34  2.78
```

```
diagnostics(est_y2)
```

```
Using `exuberdata::radf_crit2` for `cv`.
```

```
-- Diagnostics (option = gsadf) --------------------------------- Monte Carlo --

series1:  Rejects H0 at the 1% significance level
```

```
#reject null at 1%
roti = psy_ds(length(y2), rule = 1, delta = 1)
dstamp_y2 <- datestamp(est_y2, min_duration = roti)
```

```
Using `exuberdata::radf_crit2` for `cv`.
```

```
dstamp_y2
```

```
-- Datestamp (min_duration = 7) --------------------------------- Monte Carlo --

series1 :
       Start       Peak        End Duration   Signal Ongoing
1 1997-05-01 1998-04-01 1998-08-01       15 positive   FALSE
2 1998-11-01 1999-07-01 1999-10-01       11 positive   FALSE
3 1999-11-01 1999-12-01 2000-10-01       11 positive   FALSE
```

```
#cleanup
rm(PPIACO)

#gonna remove PPI, thats only in Diba and Grossman
library(onlineforecast)

#OLS
regName1 <- lm(log_price ~ log_dividend)
#regName1_2 <- ar.ols(x = regName1, order.max = 12, demean = F, intercept = T)
regname2 <- dynlm(ts(log_price) ~ L(ts(log_price)) + L(ts(log_price), 2))
#
coeftest(regName1, vcov. = vcovHC, type = "HC1")
```

```
t test of coefficients:

              Estimate Std. Error t value  Pr(>|t|)
(Intercept)  3.0208538  0.0113766  265.53 < 2.2e-16 ***
log_dividend 1.2638207  0.0047419  266.52 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
#NEXT: read on Exuber package used for GSADF from PWY

#Engle-Granger test for cointegration of prices and divs
FS_EGADF <- dynlm(ts(log_price) ~ ts(log_dividend))
FS_EGADF
```

```
Time series regression with "ts" data:
Start = 1, End = 1201

Call:
dynlm(formula = ts(log_price) ~ ts(log_dividend))

Coefficients:
     (Intercept)  ts(log_dividend)
           3.021             1.264
```

```r
# log_price = 3.02 + 1.26 * log_dividend, theta = 1.26
z_hat <- resid(FS_EGADF)
ur.df(z_hat, lags = 6, type = "none", selectlags = "AIC")
```

```
###############################################################
# Augmented Dickey-Fuller Test Unit Root / Cointegration Test #
###############################################################

The value of the test statistic is: -3.9438
```

```r
#ADF test statistic returns -3.9438, reject null hypothesis at 5%, z_hat is stationary
#critical values are -3.12, -3.41, and -3.96 respectively
#indicates log prices and log dividends cointegrated
```